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RESEARCH-ARTICLE

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ABSTRACT

A sensor node (SN) in Wireless Sensor Networks (WSN) has a limited amount of energy outfitted with a battery. Generally, sensors are positioned in such an environment wherever it is typical to reach the sensors and change or recharge them. QoS is critical in prolonging the network lifetime and attaining better throughput for a large network in WSN. Hence, a practical protocol is required that improves QoS and enhances the lifetime of the network. In this paper, QoS Learning Approach (QoS-LA) using Reinforcement Learning is developed and assessed. The proposed routing approach prolongs network lifetime by electing the optimum route to send data. The best route is selected as per extreme residual power at SN for a next-hop with excellence link and delay with error rate. The proposed approach is implemented on MATLAB on various SNs 100-1000 to judge the effectiveness on multiple parameters and assessed with state-of-the-art algorithms CNN, QoS-R, and KNN. The Simulation outcomes reveal that the QoS-LA approach offers the better results in terms of energy consumption reduces 14%-16%, number of alive nodes enhances 18%-20% that provides the network lifetime, and enhances the throughput to send data to base station 9%-11%.

KEYWORDS

WSN, Routing, Clustering, Reinforcement Learning, Cost Function, CNN, QoS-R, KNN and QoS-LA

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1 Introduction

Sensor network has a significant impact on the proficiency of civil applications, military field surveillance, and security management. Sensors are equipped with the capabilities of data communication and processing [1]. Sensors measure various conditions related to environmental surroundings and transform them into electric signals [2]. Sensors transmit sensed data via radio transmitter to the Base Station (BS) or gateway. BS performs data fusion of collected data to remove duplicity of sensed data [3]. QoS protocols in WSN have many real-time applications of target tracking and monitoring [4, 5]. Two algorithms prolong the network lifetime. The first algorithm emphasizes the whole network and other hotspot-specific areas, which tackles the difficulties of the sink node and deploys the energy adjacent to the sensor nodes. The first algorithm emphasizes the residual energy and links distance and seeks the optimum route from source to sink node. On the other hand, another algorithm emphasizes energy distribution among all SNs in the hotspot zone [8]. The cost function is used as follows:

$$f(c_{ij})=\frac{d^2}{Er_j}=\frac{(N_i-N_j)^2}{Er_j} \quad (1)$$

where Er_j represents the residual power between node j and N_i-N_j. if coverage radius be R_c h then the whole cost of route l and length D can be calculated as:

$$Pi=\sum_{i=0}^{D-1} f(c_{i,i+1}) = \sum_{i=0}^{D-1} \frac{N_{i+1}-N_i{}^2}{Er_{i+1}} \quad (2)$$

2 Related Work

An ALTA algorithm has been developed by Sun et al. [9] ALTA algorithm is built on optimization in binary of multi-objective and locates attack joining the nodes. The main idea for constructing this model covering the objective and decreasing power ingesting equilibriums the load amid the nodes minimizes time and energy cost and extends lifetime throughout the senor network. The attack detection model is used as:

$$S_{ij}[dBm]=P(d_0)[dBm]-10\gamma\log(d_{ij}/d_0)+F_{ij} \quad (3)$$

The load balances among all sensor nodes within the network can be estimated as:

$$LB=1-\sum_{j=1}^{n}(MEXT-t(B_j))\,/n\times MEXT \quad (4)$$

The appropriate symbols used in Eq. (3-4) have been summarized by the author given in [9].

Hassan et al. [10] presented a routing technique of data-centric in adaptable power where sensor node adjusts broadcast range using distance amid sink node and sender node. The author has discussed how error rate allied with the link and consistent delivery and allied energy of trustworthy transmission of packets. The total power consumed from sender to receiver among N-1 promoting nodes is:

$$E_t=\sum_{i=1}^{N}E_{i,i+1} \quad (5)$$

The power spent in the scenario of fixed power:

$$E_f=\sum_{i=1}^{N_f}\alpha R^{\gamma}=\alpha R^{\gamma}N_f \quad (6)$$

Power in variable scenario:

$$E=\sum_{i=1}^{N}\alpha D_{i,i+1}^{\gamma} \quad (7)$$

The appropriate symbols used in Eq. (5-7) have been described by author given in [10].

Yang et al. [11] proposed a novel cost function using Usage Index using the best lifetime of SNs. SUI elects sensor sets to fulfill QoS necessities and prolong the network's lifetime, and this scheme provides the most acceptable lifetime of the network. Sensors are used ideally to maximize the network's lifetime so that SNs could share all the power is ancillary QoS; therefore, QoS prerequisite and energy restraints are needed. If existence probability be P_i, $i=1\ldots\ldots N_s$ for system state i, the possible sensor sets are $N_p(i)$ named as F(i,j), $i=1\ldots N_s$, $J=1\ldots N_p(i)$, for each SN k the cost function is:

$$SUI(k)=\sum_{i=1\ldots Ns}(P(i).\sum_{k\in F(i,j):j=1..Np(i)}\frac{1}{N_p(i)}) \quad (8)$$

Mohanty et al. [12] proposed ESDAD protocol that provides redundant data after the info collecton technique applied in the transitional nodes. In the developed protocol, waiting time is calculated sagaciously for each intermediate node to collect info effectively. The sensed data by sensors transferred to the data aggregation point prudently. ESDAD calculates the cost function for hop selection and structure-free. The communication rates are adjusted during the congestion from source to transitional nodes. The cost function used in the protocol is as:

$$cfmax=\max_{i\in N}\{\alpha(E_{resd.i}+Buff_{aval.i}+l_{s.i})\} \quad (9)$$

The remaining energy of node i is estimated as:

$$E_{resd.i}=E_{level.i}-\{(E_{TX}(k,d_{tran})+E_{RX}(k))+\sum_{i=1}^{N}E_{RX}(k)+E_{agg}\} \quad (10)$$

The suitable signs used in Eq. (9-10) are described by author in [12].

An Integer Linear Program model was proposed using the cost function by Alkhdour et al. [13] to extends lifetime of network and minimize the delay. Energy is distributed among all sensor nodes to balance network load, the delay is considered as surplus objective the main objective is extending network lifetime. The cost function is used as:

$$\max\sum_{i=1}^{N}EC_i\times E_i \quad (11)$$

where EC_i is consumed energy at each sensor i while receiving the data packets and E_i is remaining energy at every sensor i.

Sheikh et al. [14] proposed a novel mechanism based on scheduling that enhances CSMA/CD using cost function as a scheduling strategy. The mechanism reduces data loss and increases throughput. The proposed mechanism is also used to grow the applications based on IoT and perform better on large network load. The packet is elected using the cost function from the queue; for that, the queue size is used. The same value is used to load the parameter in the case of two queues having an identical length. The value α=0.7 used for the longest queue so that that info could be communicated and avert the congestion. The cost function is as:

$$Cost_i=(1-\alpha)Priority_k+\alpha Load_i \quad (12)$$

Two algorithms ESCFR and DCFR proposed by Liu et al. [15] using the cost functions are as follows as:

$$TEC_{route\text{-}1}=\sum_{i,j\in route-1}EC_{ij} \quad (13)$$

$$TEC_{route\text{-}2}=\sum_{i,j\in route-2}EC_{ij} \quad (14)$$

For $TEC_{route-1} = TEC_{route-2}$ we should have

$$\sum_{i,jB\to B'} EC_{ij} = \sum_{i,jA\to A'} EC_{ij} \quad (15)$$

The appropriate symbols used in Eq. (13-15) have been particularized by the author given in [15].

Bof et al. [16] proposed an algorithm-distributed solution to grab today's applications. This algorithm achieves error buoyant evaluation in the existence of outliers and is implemented in the problem of position types. The cost function used in the developed algorithm is as:

$$\|x\|1,\epsilon := \sum_{i\epsilon\bar{S}} \sqrt{x_i^2 + \epsilon} + \sum_{i\epsilon S}(\alpha x_i^2 + \gamma) \quad (16)$$

The suitable symbols used in Eq. (1) have been explained in [16].

Assessment of the state of the art approaches is shown in Table1 [17-30].

Table 1. Assessment of existing routing approaches.

Approach	Objective	Limitations
EAP [1]	Recovers data loss, delay, throughput, and lifetime of the network	Decreases performance on large size of the network
EOSR [2]	Increases PDR, throughput and reduces power ingesting	Protocol is not entirely secured.
E-Approach [4]	Decreases energy consumption, enhances PDR and lifetime of the network.	Info confidentiality and protection ideas have not been applied.
EAC [5]	Enhances throughput and lifetime of the network, diminishes power ingesting.	GPS clustering not correctly works.
SDRP [6]	Advances security, communicational and computational performance.	Few SNs costume privacy, confidentiality protection not involved
EBRP [7]	Increases lifetime of the network reduces power ingesting, and increases throughput	Security is at the limit.
EDASRT [8]	Accomplishes better results in energy efficiency and security	Optimum value and reform interval not demarcated.
EASER [9]	Decreases power ingesting, extends the lifetime of the network, and increases PDR	EASER is not scalable
NIPA [10]	Collects info optimally, decreases power depletion and data loss	Limited security implemented
DEEC [12]	Prolongs the lifetime of the network, decreases power ingesting, advances security	In network congestion performance degrades.
SCHA [14]	Advances network correctness, recognizes malevolent nodes, offers power efficacy	High link cost
PSO [17]	Offers safe and protected surroundings, intensive care of malevolent nodes	High data aggregation cost
SEAC [19]	Recovers network security, improves network lifetime	Protection doesn't protect complete network

3 Proposed Routing Approach (QoS-LA)

In the perspective of the learning approach, every sensor is deliberated as *s* position and for every adjacent *s'* with equivalent action *(P(s'|s, a) = 1)*. Implementing action *a* at *s* shows advancing the data packet to adjacent node *s'*. The static QoS-values of *s* are stored in the routing table and provide the information of which one adjacent node will send the info. The optimum path can be created from the routing table and make a sequence of operations. Therefore, the job of learning the optimum routing approach is corresponding to discover the QoS-static points. If fundamental evolutions prototypical *P* and recompense technique *R* is recognized, the Q-static points are resolved perfectly. In maximum cases, WSNs fairly solid to regulate the fundamental model, representing the reinforcement learning approach. QoS-learning is an off-policy chronological variance (CV) rheostat algorithm that straight estimates the optimum value of the link cost function (Eq. (18)). Every period action is implemented; the mediator collects an instantaneous reward and anticipated extended-term reward to modernize QoS-values. The QoS learning approach is defined as:

$$QoS(s,a) = (1-\alpha)\ QoS(s,a) + \alpha[\gamma + \gamma \max QoS(s'a')] \quad (17)$$

Where α represents learning rate which corresponds the rate of modifying QoS values and γ represents discount factor.

A link cost function is applied on every link, which should be for the complete path. The cost function for the link between node *s* and *s'* with factors communication cost, energy stock, and error rate is calculated as:

$$Cost_{ss'} = \sum_{s=0}^{2} CF_s = CT_0 \times (Dist_{ss'})^h + CT_1 \times F(Energy_s) + CT_2 \times F(E_{ss'}) \quad (18)$$

Where $Dist_{ss'}$ represents the distance between node *s* and *s'*, *F(Energy)* is the function used present remaining power at node *s*. $F(E_s)$ function show the error rate between the link *s* and *s'*. All the three factors in the link cost function are as:

3.1 Communication Cost (($CT_0 \times (Dist_{ss'})^h$)

Where CT_0 is constant weight and h parameter depends on the environment. This factor imitates the transmission cost of power, which is proportional to the power h.

3.2 Energy Stock (($CT_1 \times F(Energy_s)$)

This factor represents residual energy of the node, the node which has higher energy for better routing.

3.3 Error Rate ($CT_2 \times F(E_{ss'})$)

Where F shows the function of nodes distance between s and s' and the buffer size on node s. the high error rate link can be avoided for network proficiency.

All the sensor nodes have real-time info creation rate α_{RT} , the total rate of real-time data generation be ρ_i nodes will be $\rho_i \alpha_{RT}$ and ε_i nodes will be added recursively to every communicating node. Then total data load of real-time can be calculated as follows:

$$\alpha_{RT}^{i} = \rho_i \alpha_{RT} + \sum_{j=1}^{\varepsilon_i} \rho_j \alpha_{RT}^{i} \tag{19}$$

4 Simulation Framework

All implementation has been completed in MATLAB version 8.1 [31] for various SNs 100-1000. SNs are extended in the sensor area 200*200 m^2. Numerous measured parameters to assess the efficacy of QoS-LA are shown in Table 2.

Table 2. Simulation Parameters

Parameters	Values
Simulation Area	200*200 m^2
Number of Nodes	100-1000
Coverage Area	200 m
Channel Type	Wireless
Distribution of Node	Random
Initial Energy	1 Joules
Packet Size(Byte)	512
Communication Channel	Bi-directional
Type of Antenna	Omni directional antenna
Radio Broadcast Model	Two way ground

5 Performance Metrics & Results Discussion

We have evaluated the performance of CNN, QoS-R, KNN and QOS-LA protocols using various metrics are as:

5.1 Energy Consumption

It signifies the entire consumption power by SNs within the network. The percentage of energy consumption can be calculated as:

$$E_C = \frac{\text{Eenergy Consumption(joules)}}{\text{Total Energy(joules)}} \times 100 \tag{20}$$

Complete energy consumption by the can be calculated as:

$$\chi_{Tx}(\beta, M) = \begin{cases} \beta\chi_{elec} + \beta\rho_{fs} M^2, M \le M_0 \\ \beta\chi_{elec} + \beta\rho_{mp} M^4, M > M_0 \end{cases} \tag{21}$$

$$\chi_{Rx}(\beta) = \beta\chi_{elec} \tag{22}$$

All the symbols and notions used in Eq. (21)-(22) are elaborated in [2].

5.2 Total alive nodes

The number of alive nodes shows the network's lifetime, and the node works until energy is drained. Percentage of total number of alive nodes within the network can be calculated as:

$$A_N = \frac{\text{Number of Alive Nodes}}{\text{Total Number of Nodes}} \times 100 \tag{23}$$

5.3 Throughput

It represents transmitted data from sender to receiver efficiently in time duration. The throughput per node within the network can be calculated as:

$$T_P = \frac{\text{Throughput at the Base Station}}{\text{Total Number of Nodes}} \tag{24}$$

QoS-LA is implemented and equated with CNN, QoS-R, and KNN. Figure 1 Fig 2 represents that QOS-LA decreases power ingesting than CNN, QoS-R, KNN. Nodes in WSN consume much energy in detecting, radio processing, and calculation when receiving and sending the info. A high volume of energy is used in searching the best and link worth route and depletion the power; typically, nodes ingest much energy at the time of receiving data than sending. Consequently, the QoS-LA protocol decreases the power consumption by 14%-16% than considered routing protocols.

Network lifetime depends on the lifetime of SNs within the network; these nodes are called alive nodes as exposed in Figure 3 & Figure 4. The network lifetime depends on how much sensor node consumes energy over time and energy availability for node use. The network's aim can be satisfied as extensively considered sensor nodes are alive and assess power utilization that forms the efficiency of the network. The proposed routing protocol QoS-LA

enhances alive node 18%-20% than CNN, QoS-R, KNN that improves network lifetime.

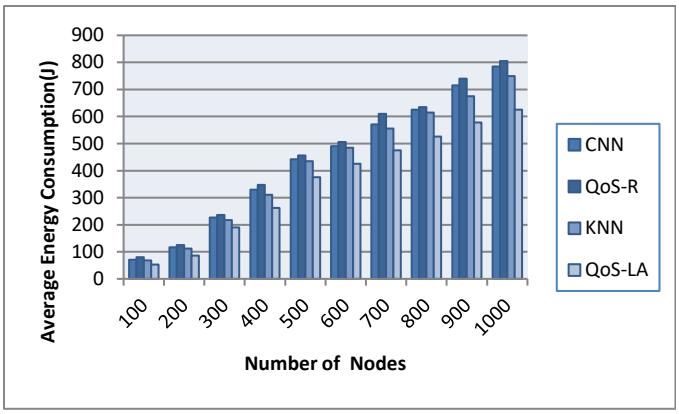

Figure 1: Average Energy Consumption (J) vs. No. of Nodes

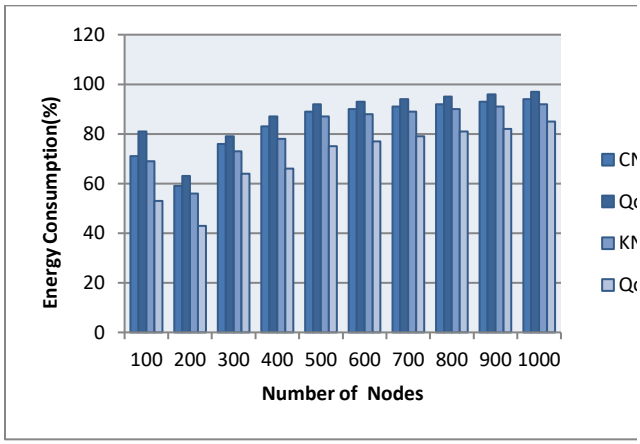

Figure 2: Energy Consumption (%) vs. No. of Nodes

In the sensor network, throughput illustrates the volume of the packet sent from sender to receiver efficiently within the time duration. Figure 5 & 6 shows that QoS-LA provides an improved throughput of 9%-11% than considered routing protocols. According to Figure 8, QoS-LA decreases end-to-end delay 13%-15% than other considered routing protocols and provides reliability for the network.

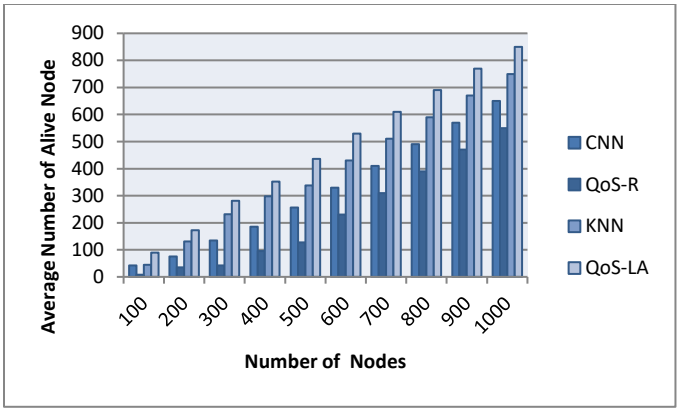

Figure 3: Average Number of Alive Nodes vs. No. of Nodes

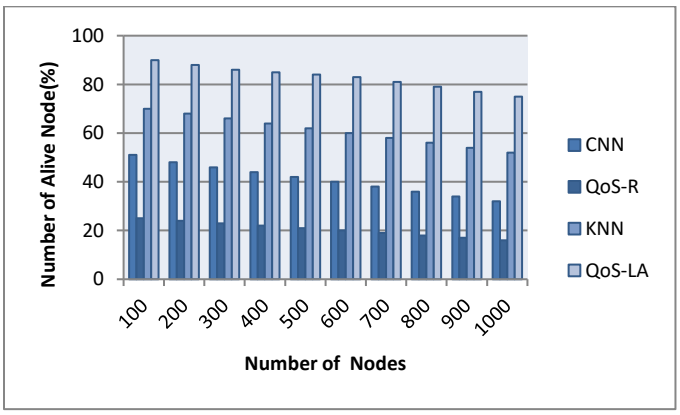

Figure 4: Number of Alive Nodes (%) vs. No. of Nodes

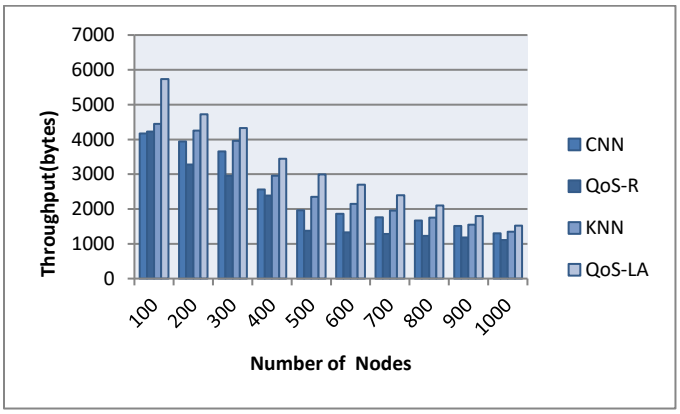

Figure 5: Throughput vs. No. of Nodes

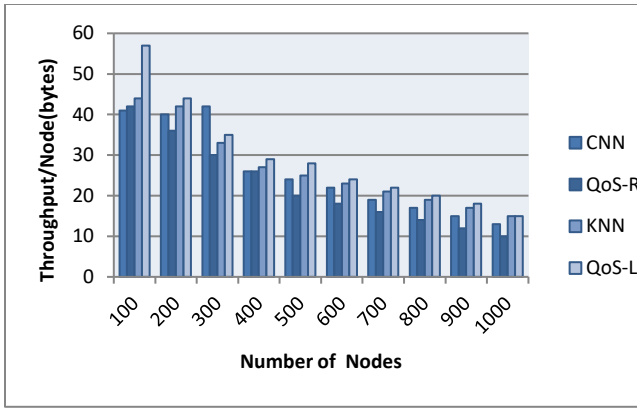

Figure 6: Throughput/Node vs. No. of Nodes

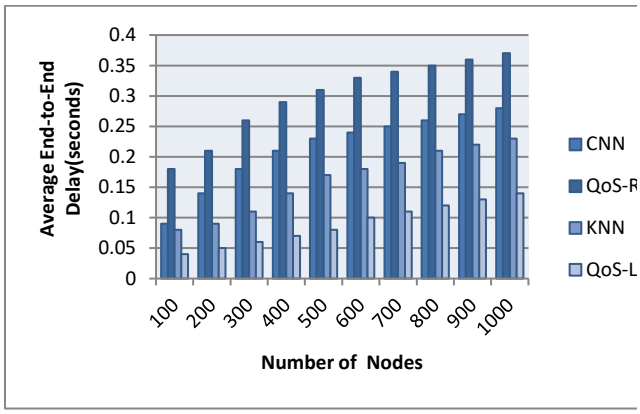

Figure 7: Everage End-to-End delay vs. No. of Nodes

6 Conclusion

In this paper, the QoS-based Energy-Aware Routing (QoS-LA) Approach is developed and assessed. Simulation results demonstrate that the QoS-LA provides better results than considered routing protocols CNN, QoS-R, KNN. QoS-LA decreases the power ingesting by 14%-16% by electing the best route amid the routes. Alive SNs are prolonged by 18%-20% that advances the lifetime of the network. The throughput is improved by 9%-11% to send data to the base station within the time. All the assessment simulation outcomes show that the QoS-LA approach is more competent than considered routing algorithms.

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